Collaborative Federated Learning Platform for Privacy-preserved Decentralized Machine Learning in Industrial Internet of Things

GROUP 08

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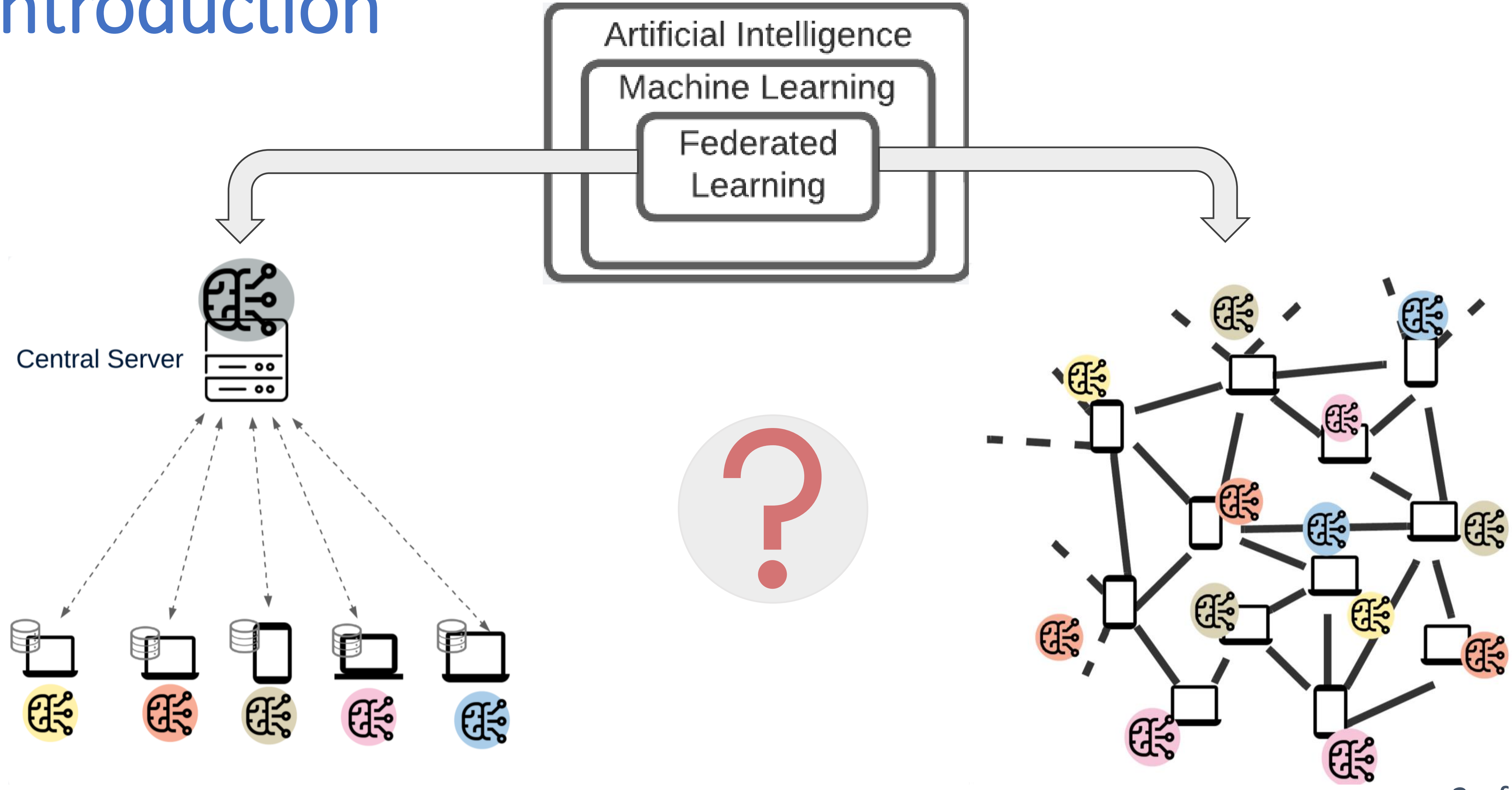
Introduction



Problem Statement

CENTRAL SERVER ATTACKS

CENTRAL SERVER FAILURES

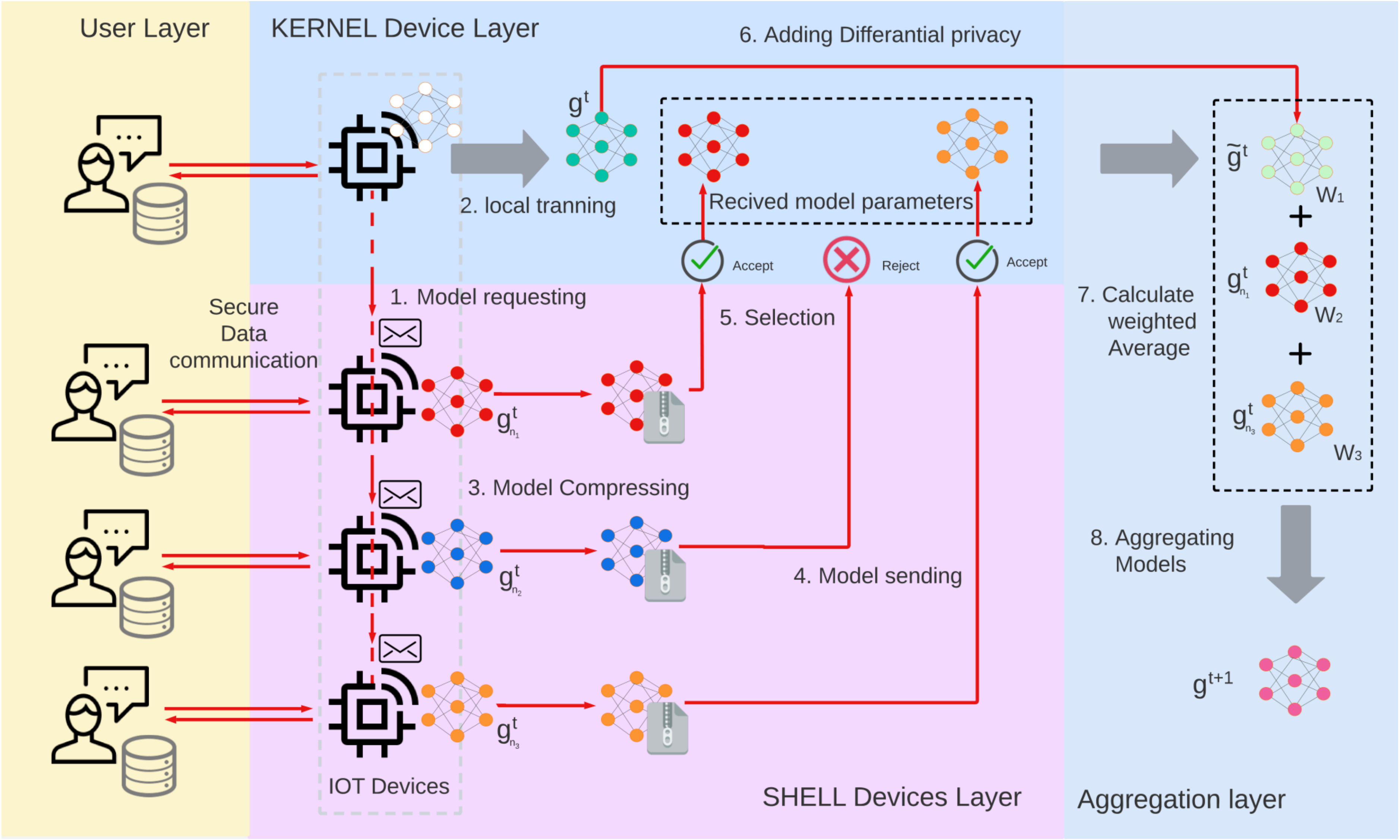
Objectives

- To comprehend the **requirement of privacy concerns** for industrial IoT devices
- To determine the solutions for **real-world IIoT applications with the FL concept**.
- To determine the tools and frameworks that can implement an FL platform.
- To design a **novel collaborative FL protocol with P2P network** architecture
- To enhance the development of **algorithms for efficient P2P communication** among supermarket applications for FL model sharing.
- To test a FL system with real-world devices in P2P network architecture.
- To implement a **self-checkout smart cart system.**
- To develop the ML algorithm for accurate predictions for the **mobile app for recommendation system**.

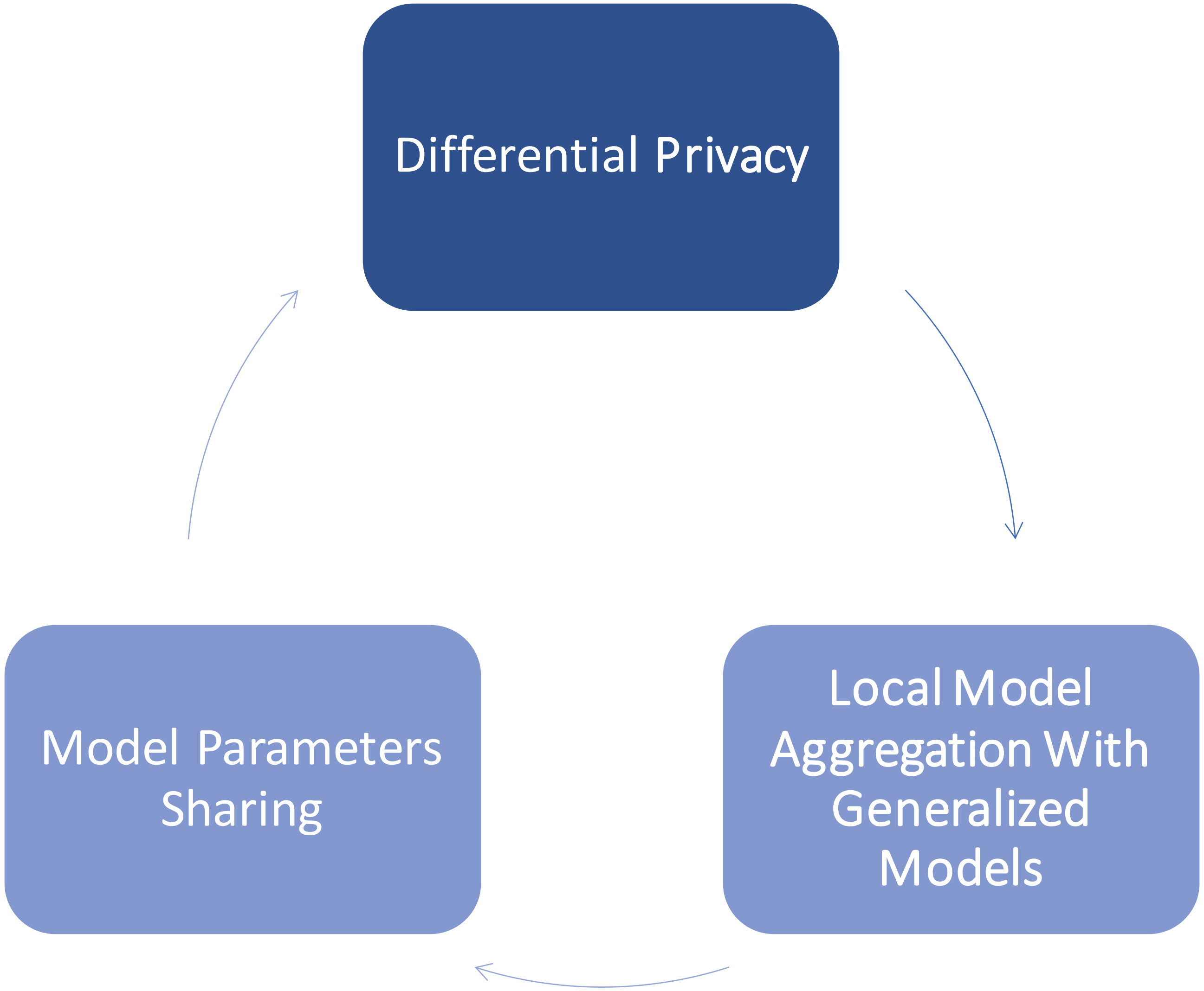
Scope

- The smart cart is modelled using a PC
- ML model in the Smart cart is locally trained after collecting a minimum required data set from check-out data
- To accept the receiving models, incoming models should be in between required range
- The supermarket app supports only Android versions higher than Android 6.0
- Ensure the data communication between the smart cart and the smartphone with SSL security.
- The existing protocols and technologies that are essential for the implementation of our novel P2P FL protocol will be utilized.

Novel Collaborative FL Platform



Privacy and Security on Platform



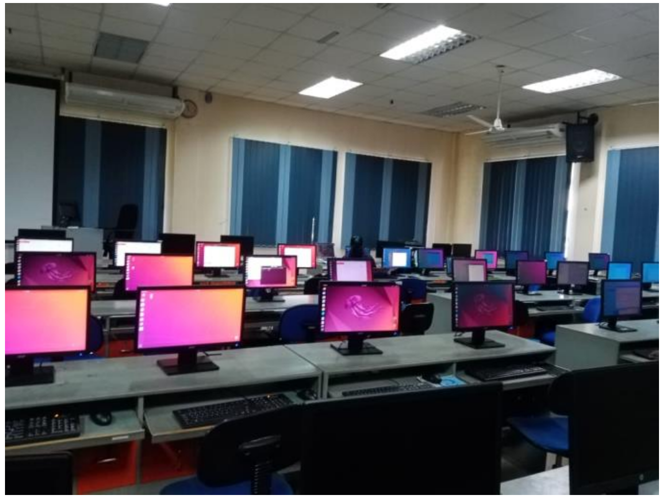
Testing Environments

1.Virtual Environments - Google Colabs , Docker

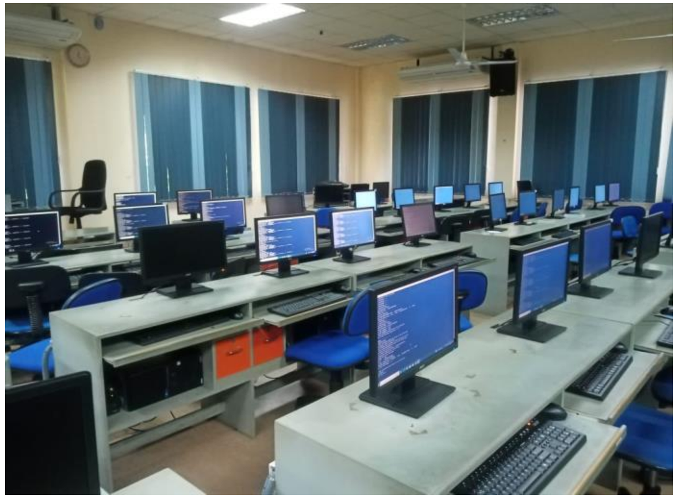
ML Model Training

P2P Network Simulation

2.Real Environments - IS Computer lab



Ubuntu Environment



Windows Environment

Focused Areas for Testing

1. **Compare Noval Protocol with Existing Protocols for**
   - Centralized Network
   - Decentralized Network

2. **Machine Learning Model Convergency**

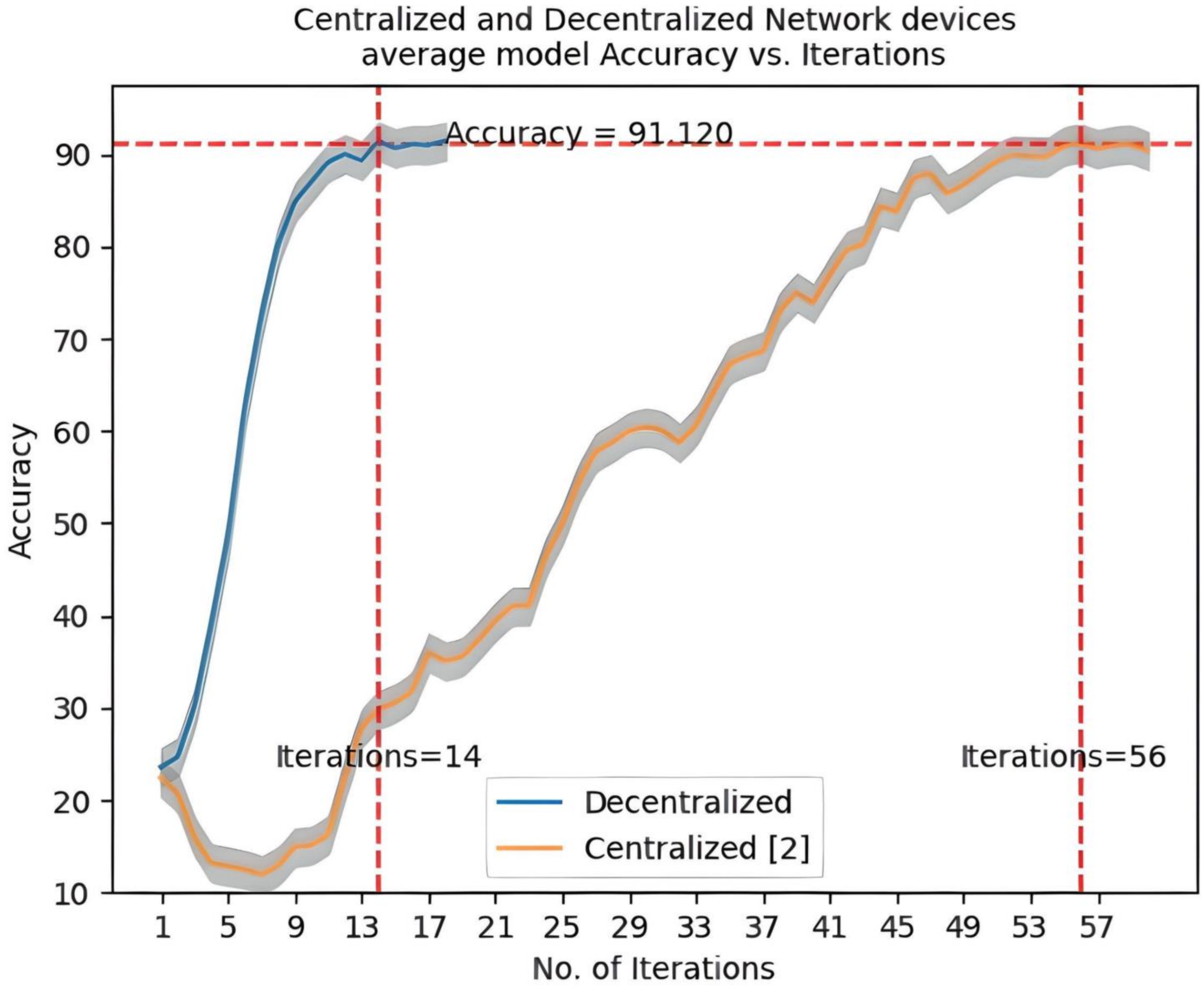
3. **Network Robustness**
   - Receiving Model Validity Analyzing
   - Differential Privacy Model

4. **Network Performance**
   - Model Receiving Time
   - Model Analyzing Time
   - Model Aggregation Time

Completed Experiments

1. Convergence Performance of Novel protocol vs Centralized FL FedAvg Protocol
2. Convergence Performance of Novel protocol vs Decentralized FL FedAvg Protocol
3. Convergence Performance of Protocol in Real Environment
4. Analyzing the Impact of Cluster Size on Aggregation Time
5. Improvement of the Kernel Time
6. Comparison of Kernel Time with number of Iterations
7. Comparison of total Received models vs Rejected models
8. Model Aggregation Count comparison with Network Convergence
9. Comparison of Model aggregation with and without differential privacy
10. Two by Two model Aggregation method comparison with existing aggregation method

Convergence Performance of Novel Protocol vs Centralized FL FedAvg Protocol



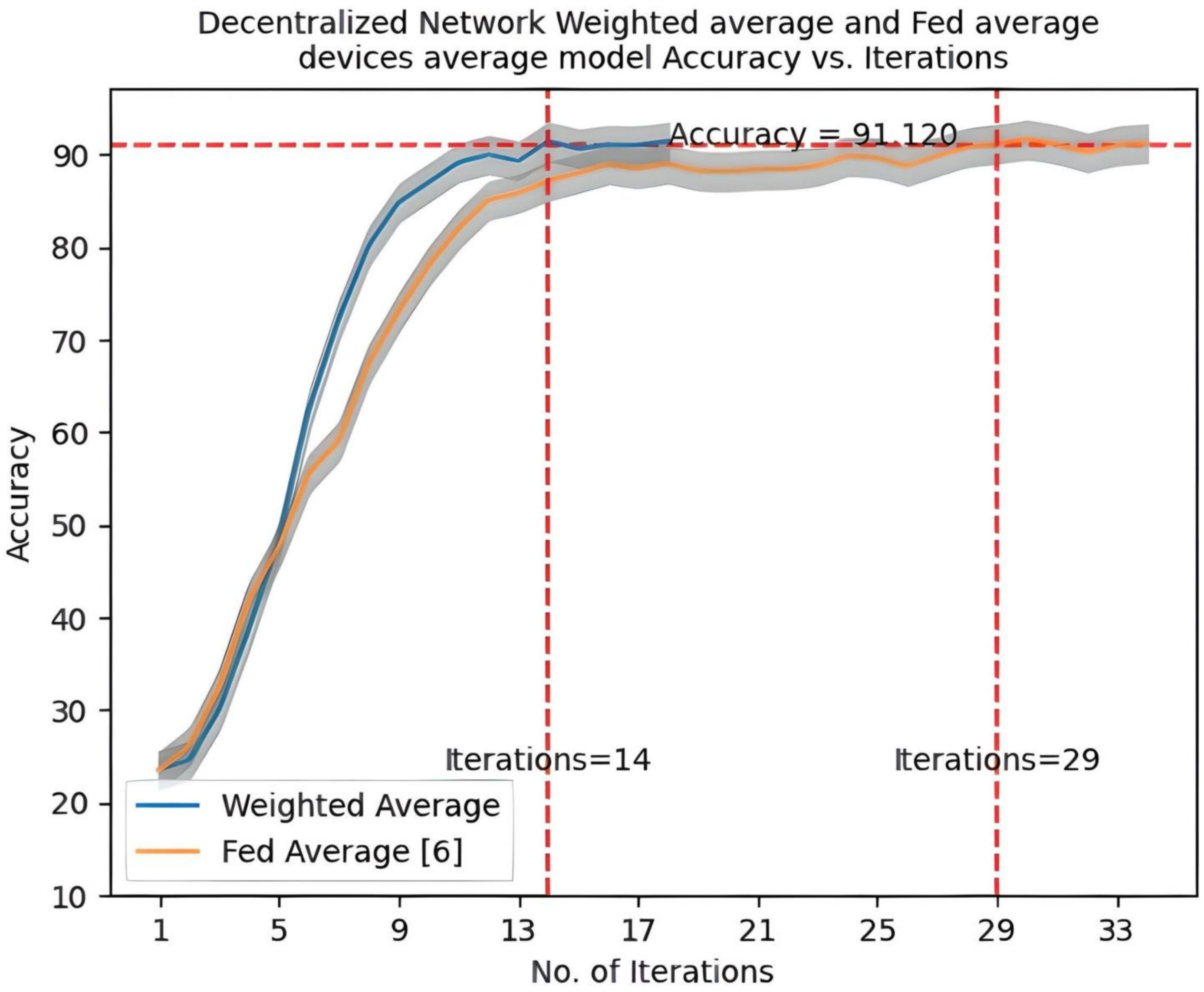
Environment

Docker Containers

I5 Computer

6 Peer Network

Convergence Performance of Novel Protocol vs Decentralized FL FedAvg Protocol



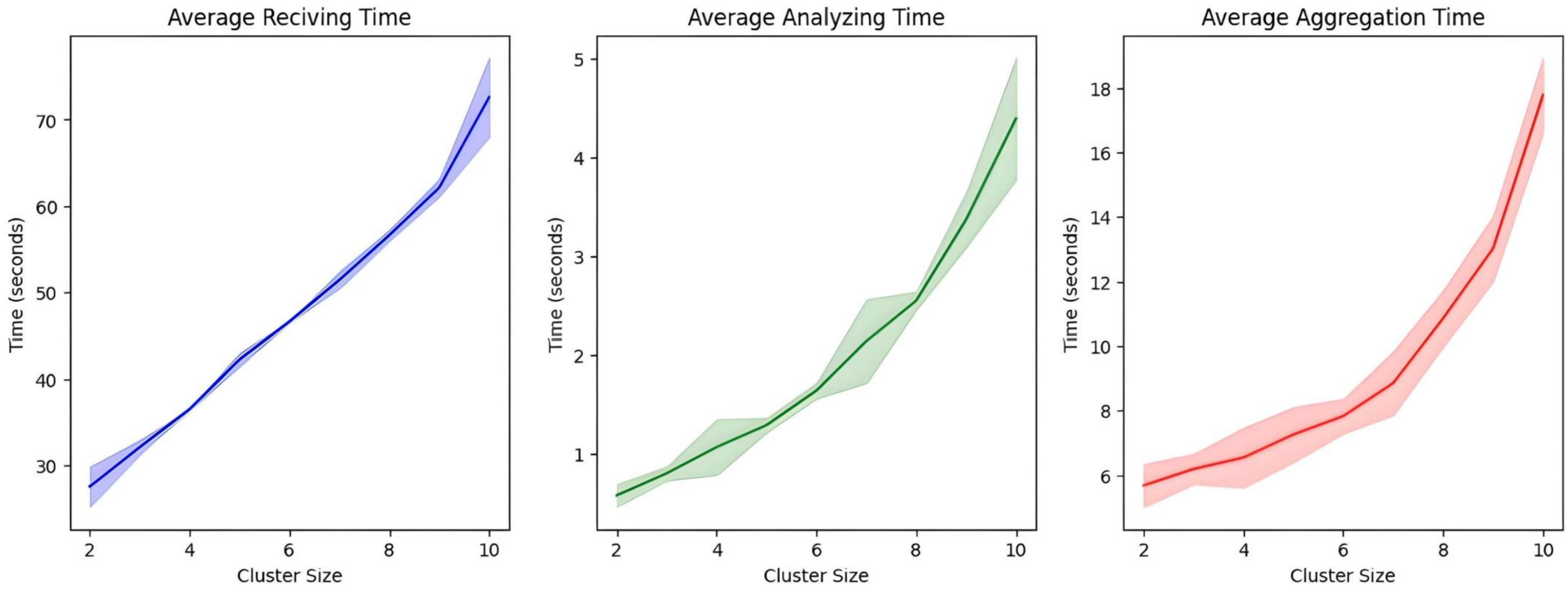
Environment

I7 Computer

Docker Containers

6 Peer Network

Analyzing the Impact of Cluster Size on Aggregation Time



Environment

Real Environment I5 Computers

25 Peer Network

Publications

Conference: IEEE Latina Conference 2023

Topic: Privacy-preserved Collaborative Federated Learning Platform for Industrial Internet of Things

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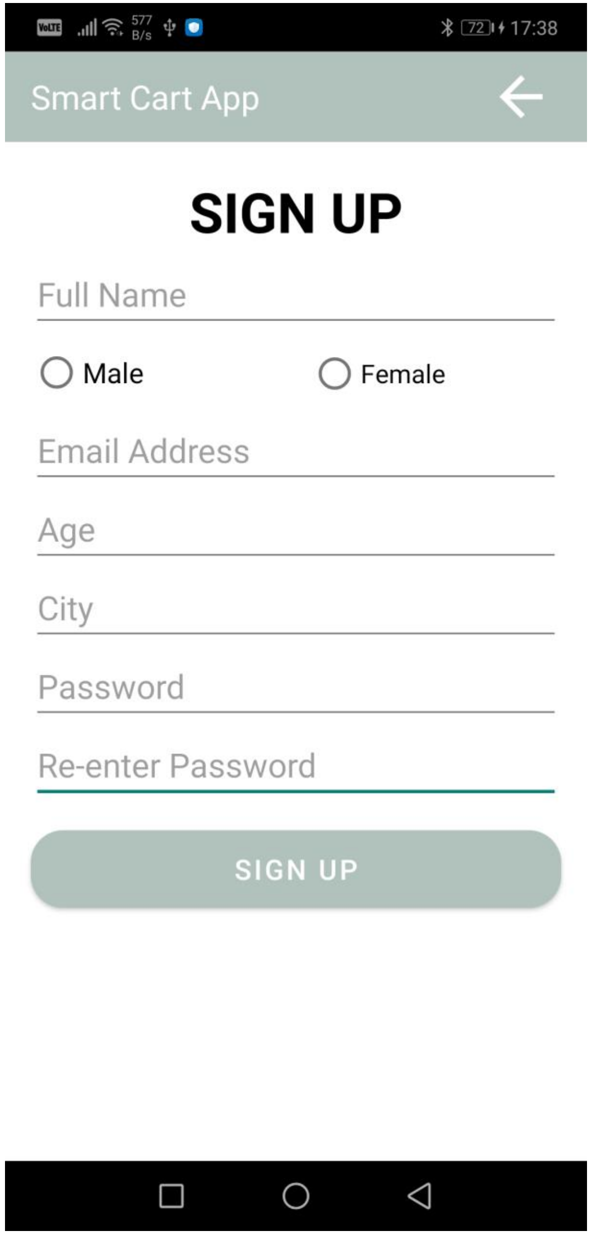
Stauts: Accepted

Product Recommendation System

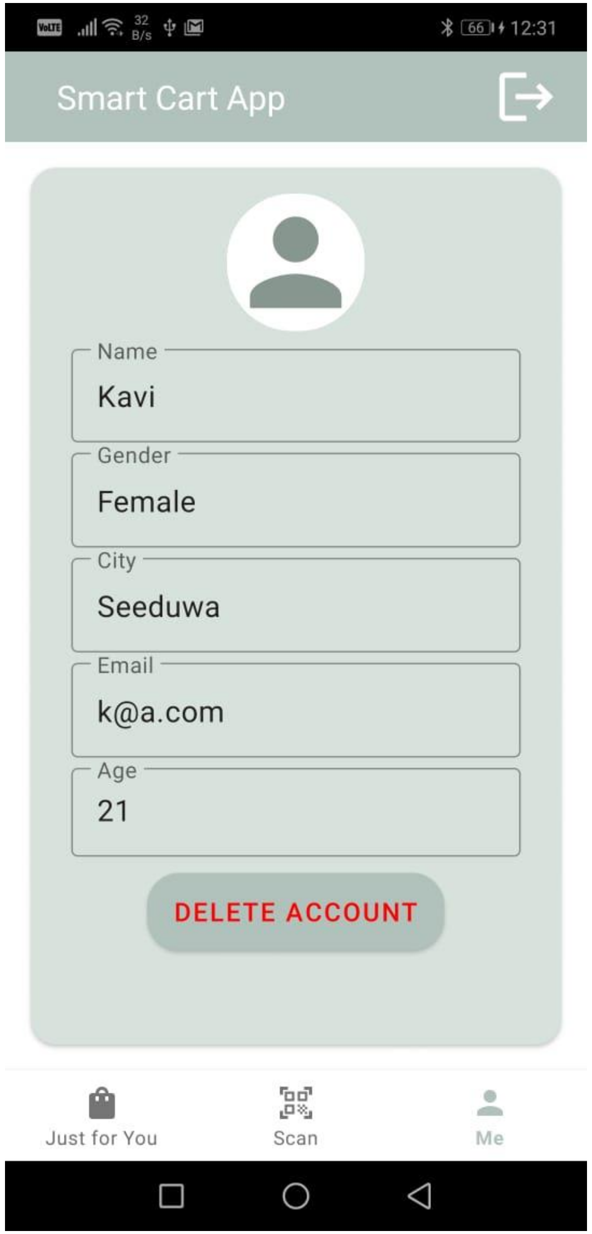
Novel Collaborative Federated Learning Protocol

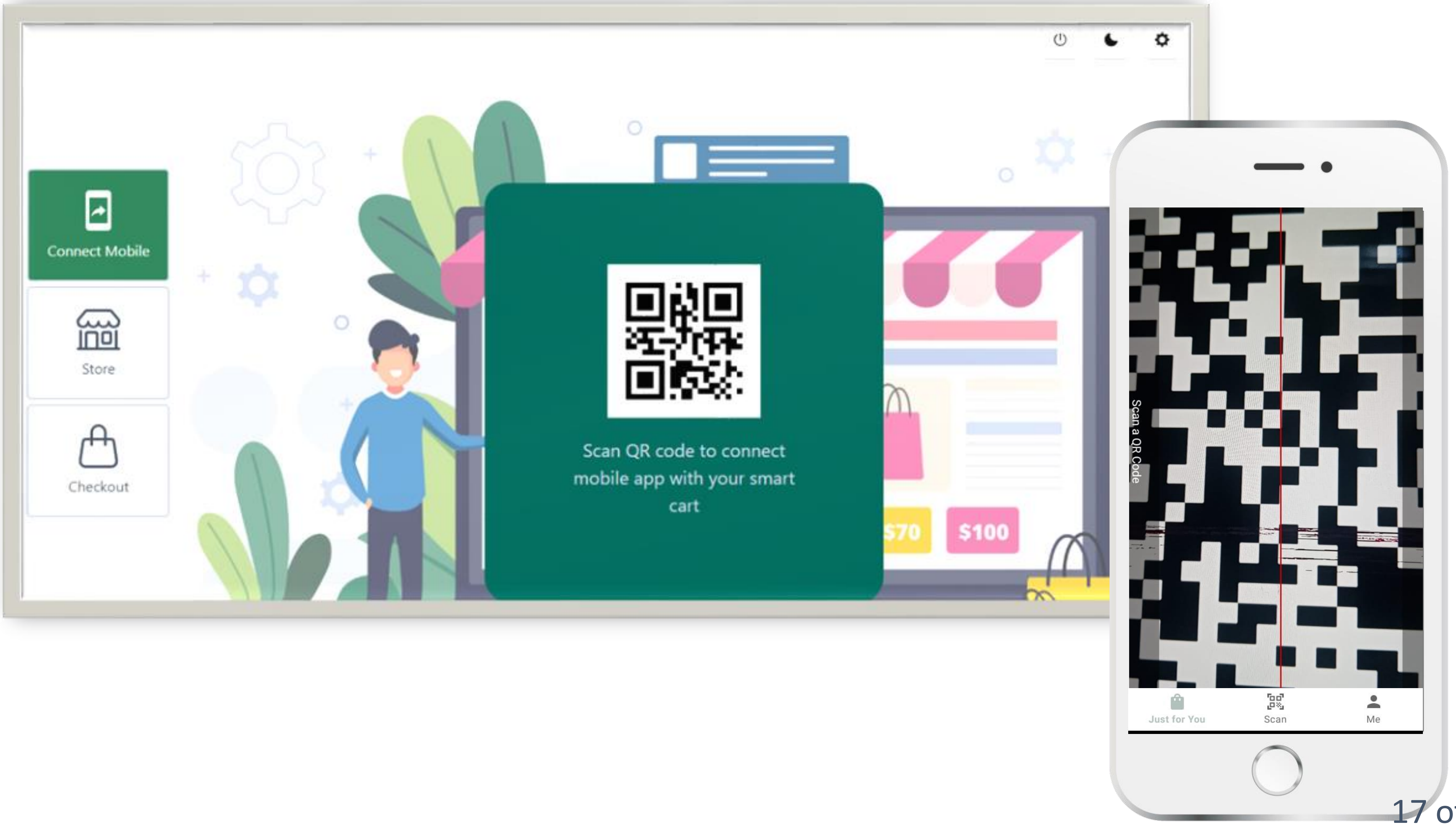
Smart Cart System

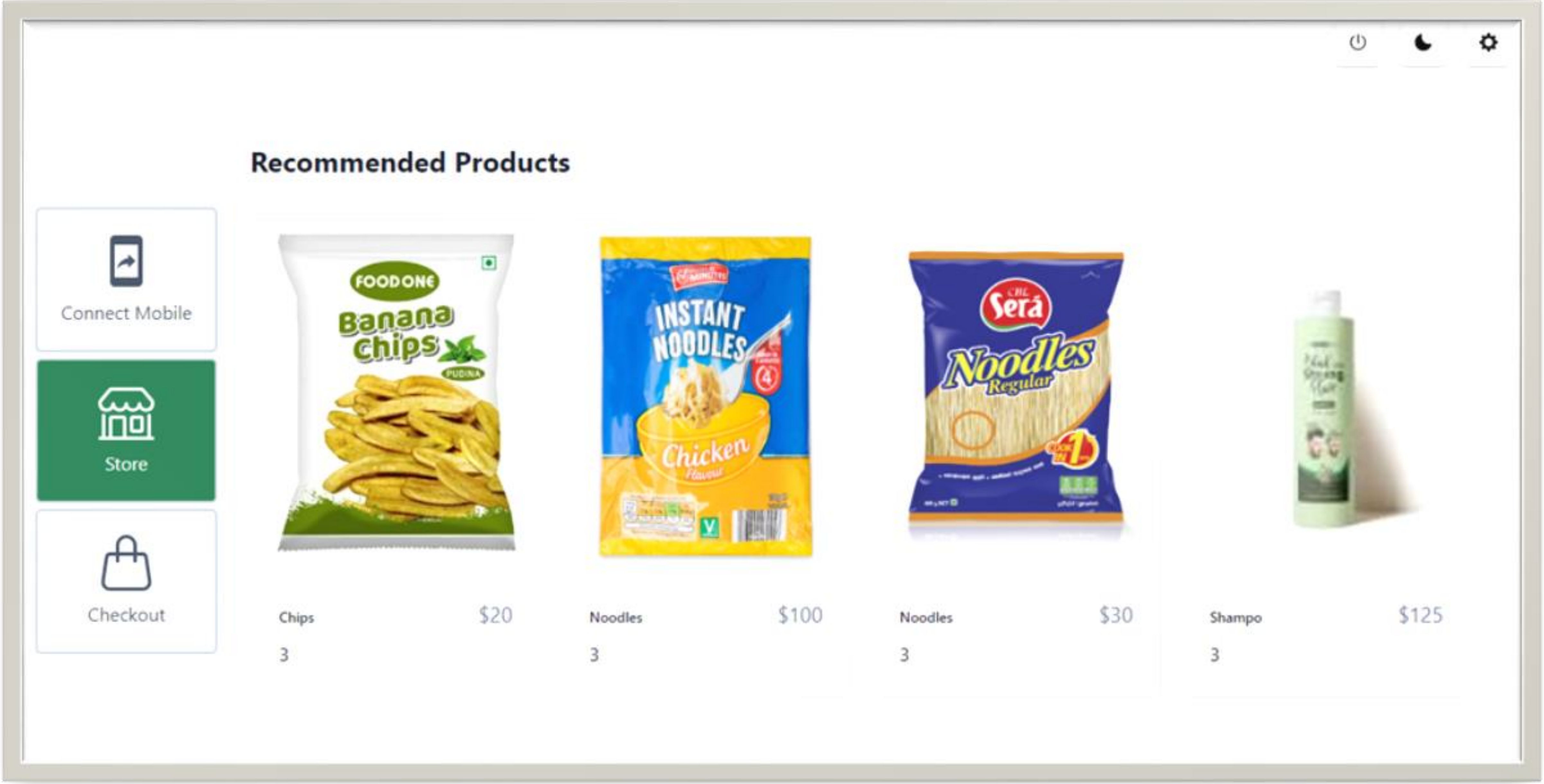
Product recommendation system (ML)

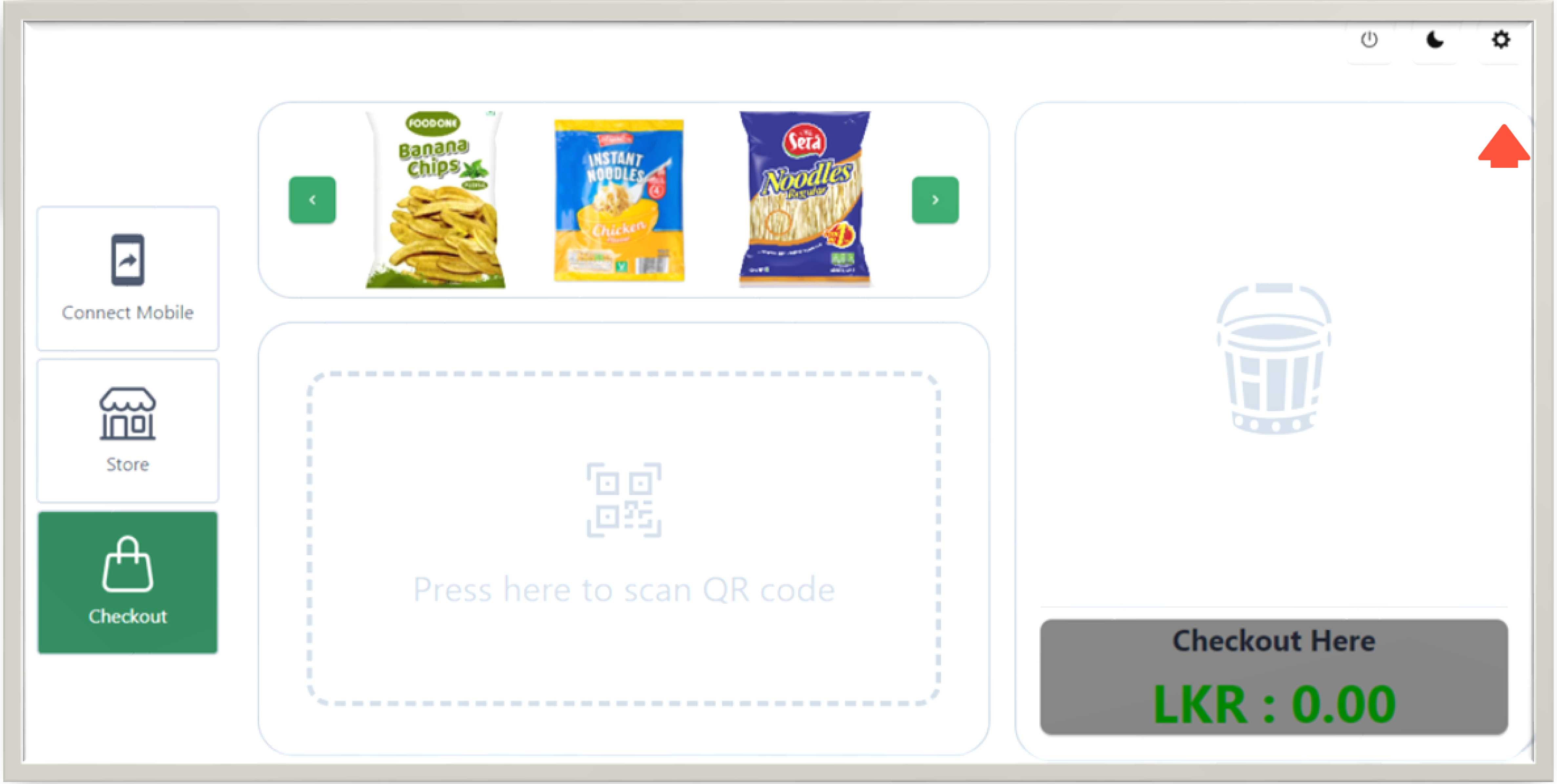

17:38
Smart Cart App
SIGN UP
Full Name
Male
Female
Email Address
Age
City
Password
Re-enter Password
SIGN UP

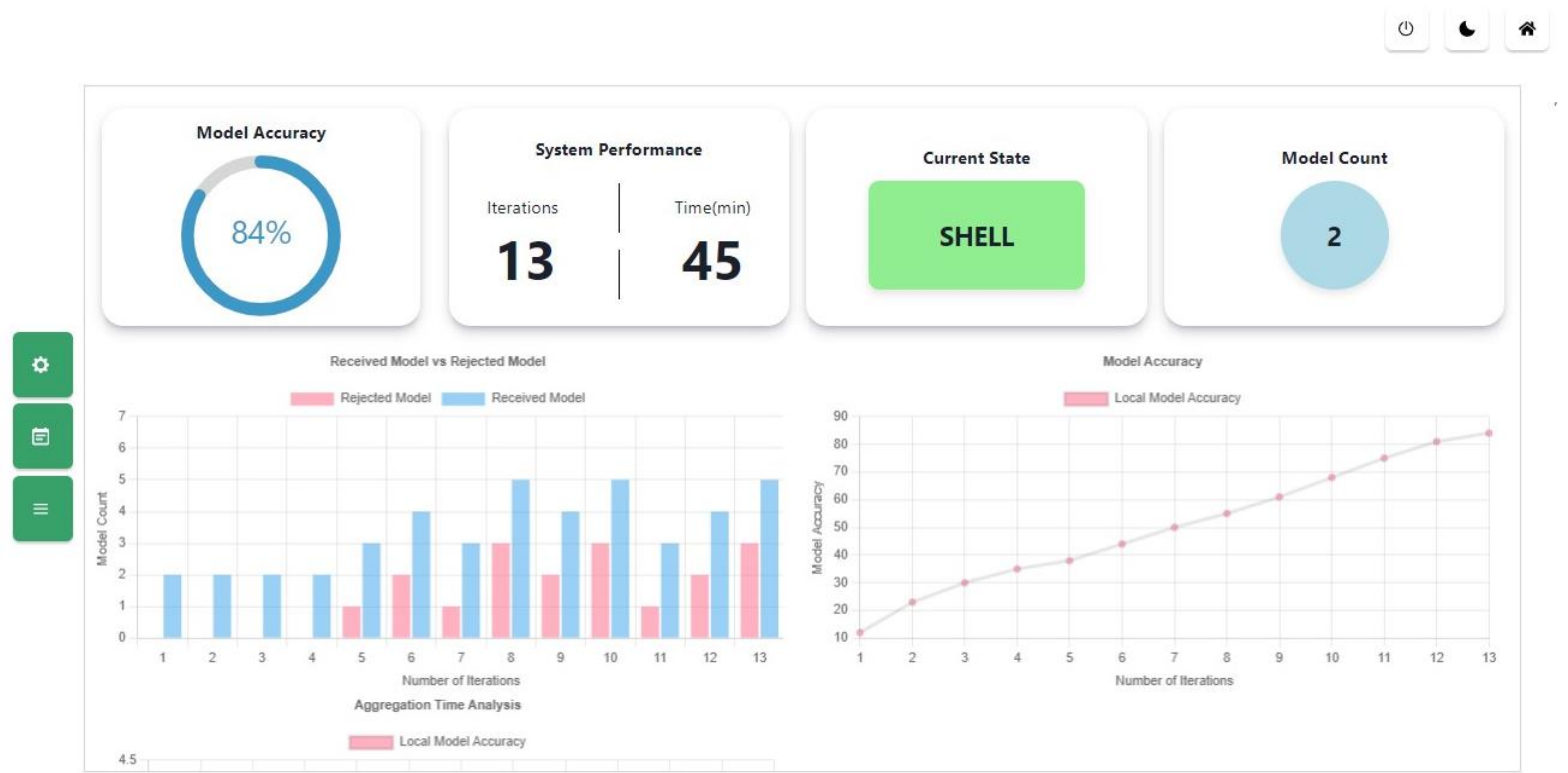

11:22
Smart Cart App
Tomatoes
Rs. 30.00
Mango
Rs. 25.00
Cabbage
Rs. 35.00
AMBEWELA
FRESH MILK
FULL CREAM
Ambewela Milk
Rs. 420.00
Just for You
Scan
Me


12:31
Smart Cart App
Name
Kavi
Gender
Female
City
Seeduwa
Email
k@a.com
Age
21
DELETE ACCOUNT
Just for You
Scan
Me


Connect Mobile
Store
Checkout
Scan QR code to connect mobile app with your smart cart
$70
$100
Scan a QR Code
Just for You
Scan
Me

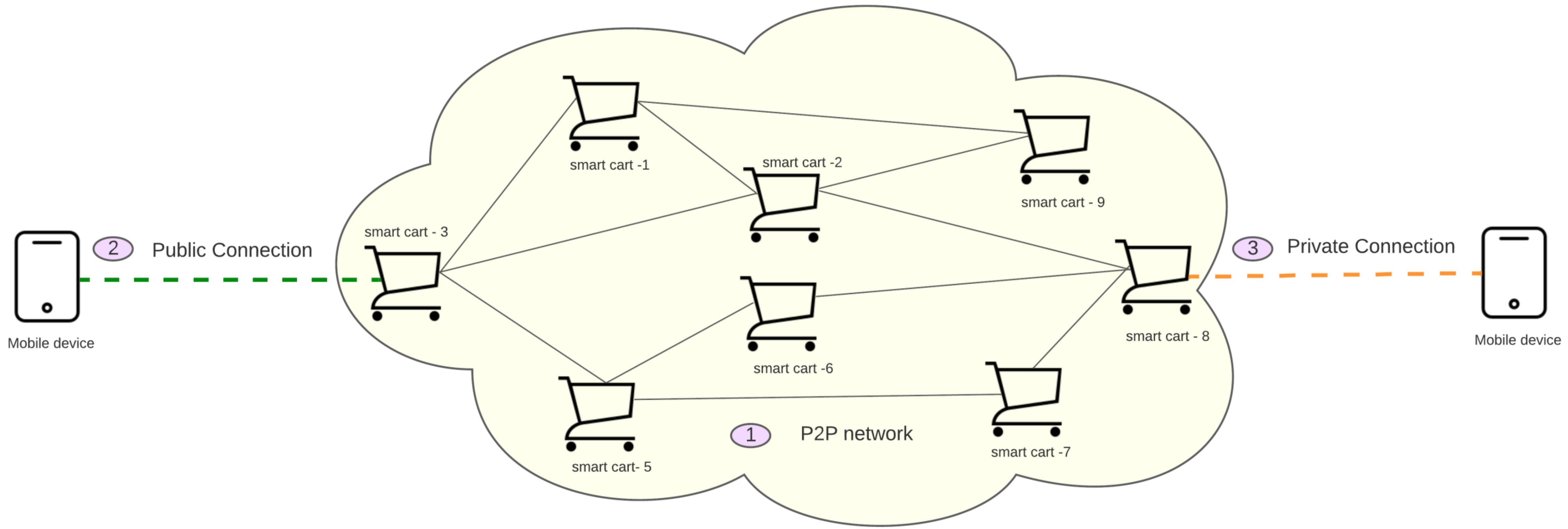

Recommended Products
Connect Mobile
Store
Checkout
FOOD ONE
Banana Chips
PUDINA
INSTANT NOODLES
Chicken
Flavour
Sera
Noodles
Regular
Chips
$20
3
Noodles
$100
3
Noodles
$30
3
Shampo
$125
3


Connect Mobile
Store
Checkout
Banana Chips
INSTANT NOODLES
Chicken
Sera
Noodles
Press here to scan QR code
Checkout Here
LKR : 0.00


Model Accuracy
84%
System Performance
Iterations
13
Time(min)
45
Current State
SHELL
Model Count
2
Received Model vs Rejected Model
Rejected Model
Received Model
Model Count
Number of Iterations
Model Accuracy
Local Model Accuracy
Model Accuracy
Number of Iterations
Aggregation Time Analysis
Local Model Accuracy


Received Model vs Rejected Model
Rejected Model
Received Model
Model Count
Number of Iterations
Model Accuracy
Local Model Accuracy
Model Accuracy
Number of Iterations
Aggregation Time Analysis
Local Model Kernal Time
Time In Minutes
Number of Iterations

Summary of the network



Thank you